EPSR-test-ex105 Pressure control by box size (control-p abc and control-p a)

Program(s): epsr | chained passes: 2 | verification: ALL IDENTICAL (all 2 passes)

What this case tests

Exercises the `control-p` option of the `epsr` engine [1,2,3], which adjusts the simulation box after every EP iteration to drive the computed pressure towards a requested value. Two runs in one folder test isotropic scaling of all three axes and scaling of only the a axis, which makes the box non-cubic.

Code exercised

Fortran (original, fortran_src/):

- `epsr_standalone_rev3.f`: `runepsr17` (`control-p` parsing, "Adjust boxsize to control pressure" block), `uniform` (`ycoh` recomputed)
- `ato_manipulation_rev4.f`: `change_boxsize`, `generate_fractional_coordinates`, `generate_cartesian_coordinates`; `press` (pressure `pressd`)
- `ato_read_and_save.f`: `minitc` (general-cell branch when the 9-number box header is read back)

Rust (port, rust/epsr-core/src/):

- `pressure.rs`: `parse`, `control_pressure`, `change_boxsize`, `generate_fractional_coordinates`
- `structure.rs`: `uniform`; `energy.rs`: `press`; `ato.rs`: `minitc`
- `epsr.rs`: `Epsr::iterate` (call after every iteration)

Physical system

The EPSR-test system: a liquid mixture of 200 dimethyl sulphoxide (DMSO, (CH3)2SO) and 400 water molecules (DMSO mole fraction 1/3; 3200 atoms) at 300 K in a cubic box of side 33.12 Å (atomic number density 0.0881 atoms/Å^3). DMSO uses EPSR's Lennard-Jones + Coulomb reference potential with methyl rotational groups; water uses SPC/E Lennard-Jones and charge parameters [4] with EPSR's flexible intramolecular geometry (O-H 0.976 Å). The empirical potential is refined against three neutron diffraction data sets of deuterated DMSO in D2O, in null-scattering H2O/D2O and in H2O (ISIS SANDALS runs SLS03863, SLS03871, SLS03850), the isotope-substitution approach of Soper and Luzar [5]. The virial pressure `pressd` computed by `press` [6] gives `rat = 0.01*clip(pressd - 0, ±1)`; the scaled axes are multiplied by `(1+rat)**(1/naxes)`, molecule centres are transformed through fractional coordinates of the old cell, and `dmax`/`cell` are reset to the smallest perpendicular face distance.

Inputs

`scripts/make_inputs.sh`: in the EPSR-test `.EPSR.inp` `control-p abc 0.0 0.01`, `ntimes 2`, `niter 2`; `scripts/clone_run.py boxa "control-p a 0.0 0.01"` makes a second run `boxa` with the same files. `commands.txt`: both runs, `repeat 2`.

How the accuracy was verified

`commands.txt` (2 program calls, repeated 2 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; only the date line of `.out` and the wall-clock columns of `fort.35` are masked). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 2 passes.
Four iterations per run (2 passes × `niter 2`). In the stored Fortran reference the cubic run grows from 33.1213 to 33.4521 Å; in `boxa` only a changes (34.3514 × 33.1213 × 33.1213 Å), the `.ato` is written with the general-cell header and read back in pass 2. The README reports gcov evidence that the `dmaxfac`, `change_boxsize` and `uniform` lines run in every iteration. Both runs' outputs are identical in both passes.

Notes

Reproduced: `rho` and `xstar` are not recomputed after a box change (so `uniform` gives the same `ycoh`); phix/phiy/phiz of all molecules are transformed; the keyword `ca` is accepted but selects no axis and `ac` is rejected.

References

[1] A. K. Soper, "Empirical potential Monte Carlo simulation of fluid structure", Chem. Phys. 202, 295-306 (1996).
[2] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[3] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[4] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).
[5] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[6] M. P. Allen and D. J. Tildesley, Computer Simulation of Liquids, Oxford University Press (1987).
[7] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.